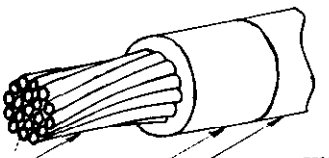
FED. SUP CLASS
6145



NICKEL-COATED COPPER CONDUCTOR

POLYIMIDE, DIP-COATED, 0.0007 ± 0.0002 in.

EXTRUDED POLYTETRAFLUOROETHYLENE (TFE)

© ⓓ CONSTRUCTION DETAILS ©

MS PART NO.	WIRE SIZE	STRANDING	MAX. DIA. STRANDED COND. (INCH)	FINISHED WIRE		
				MAX. RESISTANCE AT 20°C (68°F)-CLASS 11 (OHMS/1000 FT.)	DIAMETER (INCH)	MAX. WEIGHT LBS./1000 FT.
MS18105-28-*	28	7 X 36	0.016	67.6	0.035±.002	1.30
MS18105-26-*	26	19 X 38	0.021	41.0	0.039±.002	1.80
MS18105-24-*	24	19 X 36	0.026	25.1	0.043±.002	2.60
MS18105-22-*	22	19 X 34	0.033	15.7	0.050±.002	3.60
MS18105-20-*	20	19 X 32	0.041	9.67	0.058±.002	5.40
MS18105-18-*	18	19 X 30	.051	6.03	0.070±.002	7.90
MS18105-16-*	16	19 X 29	.058	4.73	0.080±.002	10.30
MS18105-14-*	14	19 X 27	.073	2.99	0.100±.003	17.38
MS18105-12-*	12	37 X 28	.090	1.97	0.120±.004	26.00

PERFORMANCE DETAILS

MS PART NO.	ABRASION RESISTANCE (PROCEDURE II)		WEIGHT (LBS.)	TENSION LOAD (LBS.)	BEND TESTING				
					MANDREL DIAMETER (INCHES-MAX.)			TEST LOAD (POUNDS)	
	MINIMUM INCHES OF TAPE	WEIGHT SUPPORT BRACKET			LIFE CYCLE TEST	COLD BEND TEST	WRAP TEST	LIFE CYCLE TEST	COLD BEND TEST
MS18105-28-*	15	A	0.50	1.00	1/8	1/8	1/4	0.50	0.50
MS18105-26-*	21	A	0.50	1.00	1/8	1/8	1/4	0.50	0.50
MS18105-24-*	21	A	0.50	1.00	1/8	1/8	1/4	0.50	0.50
MS18105-22-*	30	A	0.50	1.00	1/4	1/4	1/4	0.75	0.75
MS18105-20-*	30	A	0.50	1.00	1/4	1/4	1/4	0.75	0.75
MS18105-18-*	24	B	0.75	1.00	1/4	1/4	3/8	2.00	2.00
MS18105-16-*	24	B	0.75	2.00	1/4	1/4	3/8	2.00	2.00
MS18105-14-*	36	B	1.25	2.00	3/4	3/4	3/4	3.00	3.00
MS18105-12-*	36	B	1.25	2.00	3/4	3/4	3/4	4.00	4.00

TEMPERATURE RATING: 260°C (500°F) Maximum conductor temperature
VOLTAGE RATING: Sizes 28-16 inclusive, 600 volts; Sizes 14-12, 1000 volts.
INSULATION FLAWS: 60 CY RMS-5000 Volts (Minimum)
SHRINKAGE: 0.03 Inch (Max.) in 12 Inches @275°C
BLOCKING: 260°C (500°F)
SMOKE: 275°C (527°F)
COLD BEND: 2 Turns (720°)
DIELECTRIC: 3000 Volts - 60 CY RMS
DIELECTRIC TANK TEST: 2200 Volts - 60 CY RMS, **Sizes 14-12, 3000 volts 60 CY RMS**
INSULATION RESISTANCE: 50,000 Megohms Per 1000 Ft. (Minimum) 100% test performed during Diel. Tank Test
AIR OVEN: 275°C ± 3°C (527°F ± 5°F)
SURFACE RESISTANCE: 500 Megohms—In.Min.
COLOR: Yellow preferred. Available also in the following solid colors: Red, Orange, Green, Black, and Brown. All Colors shall be in accordance with Mil-Std-104, class 1. White not available because of the nature of the polyimide coating.
IDENTIFICATION OF PRODUCT: Required (#28-#22 Print Vertical) **(Sizes 20-12, print vertical or horizontal).**
MS PART NO.: The * shown in the tables above shall be replaced by a color code designator as shown below:

YELLOW	RED	ORANGE	GREEN	BROWN	BLACK
4	2	3	5	1	0

EXAMPLE: Size 20, Yellow: MS18105-20-4
Red: MS18105-20-2.

© CANCELLED AFTER 20 SEPTEMBER 1968, NO SUPERSEDING DOCUMENT.

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 10 March 1964 REVISED (A) 21 September 1964 (B) 7 December 1964 © 20 SEPT 1968

This standard has been approved by the NAVAL AIR SYSTEMS COMMAND Department of the NAVY and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY- AS Other Cust	TITLE WIRE, ELECTRIC, EXTRUDED-TFE-FLUOROCARBON INSULATED, POLYIMIDE-COATED, NICKEL-COATED COPPER.	MILITARY STANDARD MS 18105 (AS)
PROCUREMENT SPECIFICATION MIL-W-22759(NEP)	SUPERSEDES:	SHEET OF

DD FORM (SEP 67) 672-1 (Limited coordination)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-341-513/A-2562

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